Sources: Sheldon Axler, *Measure, Integration & Real Analysis* (MIRA) and its *Supplement*, at measure.axler.net.

Lemma 1. Suppose $a < b$ and $u\colon \mathbf{R} \to \mathbf{R}$ is Borel measurable with $u|_{[a,b]}$ continuous. Then $\int_a^b u(x)\,dx = \int \chi_{[a,b]} u\,d\lambda$, where the left side is the Riemann integral.

Proof. The restriction $u|_{[a,b]}$ is bounded, because a continuous function on a closed bounded set is bounded, and it is Riemann integrable because continuous functions are Riemann integrable (1.11). Thus the equality of the Riemann and Lebesgue integrals (3.34) gives $\int_a^b u(x)\,dx = \int_{[a,b]} u\,d\lambda$, and the right side equals $\int \chi_{[a,b]} u\,d\lambda$, because by integrals via finite simple functions (3.9), applied to u^+ and u^-, both are the supremum of the same sums $\sum_{j=1}^m c_j \lambda(A_j)$, the sets A_j with $c_j > 0$ lying in $[a,b]$ in both cases. □

Exercise 3.1 (MIRA 3B.14). Let λ denote Lebesgue measure on $\mathbf{R}$.

(a) Let $f(x) = 1/\sqrt{x}$. Prove that $\int_{[0,1]} f\,d\lambda = 2$.

(b) Let $f(x) = 1/(1+x^2)$. Prove that $\int_{\mathbf{R}} f\,d\lambda = \pi$.

(c) Let $f(x) = (\sin x)/x$. Show that the integral $\int_{(0,\infty)} f\,d\lambda$ is not defined but $\lim_{t\to\infty} \int_{(0,t)} f\,d\lambda$ exists in $\mathbf{R}$.

Solution 3.1. (a) Define $f\colon \mathbf{R} \to [0,\infty)$ by $f(x) = 1/\sqrt{x}$ for $x > 0$ and $f(x) = 0$ for $x \le 0$, which agrees with Example 1.15 on $[0,1]$. Then f is Borel measurable, because $f|_{(0,\infty)}$ is continuous (every continuous function is Borel measurable, 2.41) and f is 0 on $(-\infty, 0]$. By the definition of integration on a subset (3.24), $\int_{[0,1]} f\,d\lambda = \int \chi_{[0,1]} f\,d\lambda$. For $j \in \mathbf{Z}^+$ let $g_j = \chi_{[1/j,1]} f$, which is nonnegative and measurable. Because $f \ge 0$ and $[1/j, 1] \subseteq [1/(j+1), 1]$, we have $0 \le g_1 \le g_2 \le \cdots$, and $\lim_{j\to\infty} g_j(x) = \chi_{[0,1]}(x) f(x)$ for every $x \in \mathbf{R}$: if $x \in (0,1]$, then $g_j(x) = f(x)$ for all $j \ge 1/x$, and otherwise both sides are 0 because $f(0) = 0$ and $\chi_{[0,1]}(x) = 0$ for $x \notin [0,1]$. Thus the Monotone Convergence Theorem (3.11), Lemma 1 applied to f, which is continuous on $[1/j, 1]$, and the fundamental theorem of calculus give

$$\int_{[0,1]} f\,d\lambda = \lim_{j\to\infty} \int \chi_{[1/j,1]} f\,d\lambda = \lim_{j\to\infty} \int_{1/j}^1 \frac{dx}{\sqrt{x}} = \lim_{j\to\infty} \left(2 - \frac{2}{\sqrt{j}}\right) = 2.$$

(b) The function $f(x) = 1/(1+x^2)$ is continuous on $\mathbf{R}$, hence Borel measurable, and $f > 0$. By 3.24, $\int_{\mathbf{R}} f\,d\lambda = \int f\,d\lambda$. For $j \in \mathbf{Z}^+$ let $f_j = \chi_{[-j,j]} f$. Then $f_1, f_2, \ldots$ are nonnegative and measurable with $f_1 \le f_2 \le \cdots$, and $\lim_{j\to\infty} f_j(x) = f(x)$ for every $x \in \mathbf{R}$ because $x \in [-j, j]$ for all $j \ge |x|$. Thus 3.11 and Lemma 1 (f is continuous on $[-j, j]$) give

$$\int_{\mathbf{R}} f\,d\lambda = \lim_{j\to\infty} \int_{-j}^{j} \frac{dx}{1+x^2} = \lim_{j\to\infty} \big(\arctan j - \arctan(-j)\big) = \lim_{j\to\infty} 2\arctan j = \pi,$$

because $\arctan$ is an odd antiderivative of $x \mapsto 1/(1+x^2)$ and $\lim_{t\to\infty} \arctan t = \pi/2$.

(c) Now $f(x) = (\sin x)/x$ is continuous on $(0,\infty)$. Define $g\colon \mathbf{R} \to \mathbf{R}$ by $g(x) = f(x)$ for $x > 0$ and $g(x) = 0$ for $x \le 0$; then g is Borel measurable as in (a), and $|g| \le 1$ because $|\sin x| \le x$ for $x > 0$. Because g extends f by 0, 3.24 gives $\int_{(0,\infty)} f\,d\lambda = \int_{(0,\infty)} g\,d\lambda = \int \chi_{(0,\infty)} g\,d\lambda = \int g\,d\lambda$, each side defined exactly when the next is.

First we show that $\int g\,d\lambda$ is not defined. For $i \in \mathbf{Z}^+$ let $I_i = [i\pi, (i+1)\pi]$; on I_i the sign of $\sin x$ is that of $(-1)^i$, and $1/x \ge 1/((i+1)\pi)$. Suppose $n \in \mathbf{Z}^+$. The intervals $I_2, I_4, \ldots, I_{2n}$ are

disjoint, and thus the measurable function $\varphi_n(x) = \sum_{m=1}^n \chi_{I_{2m}}(x)(\sin x)/\big((2m+1)\pi\big)$ satisfies $0 \le \varphi_n \le g^+$ on $\mathbf{R}$. The mth term is Borel measurable and continuous on I_{2m}, and thus Lemma 1 and the fundamental theorem of calculus give

$$\int \chi_{I_{2m}} \frac{\sin x}{(2m+1)\pi}\, d\lambda = \int_{2m\pi}^{(2m+1)\pi} \frac{\sin x}{(2m+1)\pi}\, dx = \frac{2}{(2m+1)\pi}.$$

Hence integration is order preserving (3.8) and additivity of integration (3.16) give

$$\int g^+\, d\lambda \ge \int \varphi_n\, d\lambda = \sum_{m=1}^n \frac{2}{(2m+1)\pi} \ge \frac{1}{\pi} \sum_{k=2}^{n+1} \frac{1}{k} = \frac{s_{n+1} - 1}{\pi},$$

where $s_N = \sum_{k=1}^N 1/k$ and $2/(2m+1) \ge 1/(m+1)$. Because $s_{2^{M+1}} - s_{2^M}$ is a sum of 2^M terms, each at least 2^{-M-1}, we have $s_{2^M} \ge 1 + M/2$ for every integer $M \ge 0$. Taking $n = 2^M - 1$ gives $\int g^+\, d\lambda \ge M/(2\pi)$ for every $M \in \mathbf{Z}^+$; hence $\int g^+\, d\lambda = \infty$. In the same way, the intervals $I_1, I_3, \ldots, I_{2n-1}$ are disjoint, on I_{2m-1} we have $g^-(x) = -g(x) \ge (-\sin x)/(2m\pi) \ge 0$, and the measurable function $\psi_n(x) = \sum_{m=1}^n \chi_{I_{2m-1}}(x)(-\sin x)/(2m\pi)$ satisfies $0 \le \psi_n \le g^-$ on $\mathbf{R}$. By Lemma 1,

$$\int \chi_{I_{2m-1}} \frac{-\sin x}{2m\pi}\, d\lambda = \int_{(2m-1)\pi}^{2m\pi} \frac{-\sin x}{2m\pi}\, dx = \frac{2}{2m\pi}.$$

Thus 3.8 and 3.16 give

$$\int g^-\, d\lambda \ge \int \psi_n\, d\lambda = \sum_{m=1}^n \frac{2}{2m\pi} = \frac{s_n}{\pi},$$

which for $n = 2^M$ is at least $(1 + M/2)/\pi$. Thus $\int g^-\, d\lambda = \infty$ as well, and $\int g\, d\lambda$ is not defined by the definition of the integral of a real-valued function (3.18); hence neither is $\int_{(0,\infty)} f\, d\lambda$.

Now let $F(t) = \int \chi_{(0,t)} g\, d\lambda$ for $t > 0$. If $A \subseteq [0,t]$ is a Borel set, then $|\chi_A g| \le \chi_A$, and thus 3.8 and the integral of a characteristic function (3.4) give $\int |\chi_A g|\, d\lambda \le \lambda(A) \le \lambda([0,t]) = t$; hence $\chi_A g \in \mathcal{L}^1(\lambda)$. In particular $F(t) \in \mathbf{R}$, and because $g|_{(0,t)} = f|_{(0,t)}$, 3.24 gives $\int_{(0,t)} f\, d\lambda = F(t)$.

Define $h, q\colon \mathbf{R} \to \mathbf{R}$ by $h(x) = (\cos x)/x^2$ and $q(x) = 1/x^2$ for $x \ge 1$ and $h(x) = q(x) = 0$ for $x < 1$. Both are Borel measurable as in (a), and $|h| \le q$. Because $0 \le \chi_{[1,2]} q \le \chi_{[1,3]} q \le \cdots$ and $\lim_{s\to\infty} \chi_{[1,s]} q = q$ pointwise, 3.11 and Lemma 1 (q is continuous on $[1,s]$) give

$$\int q\, d\lambda = \lim_{s\to\infty} \int_1^s \frac{dx}{x^2} = \lim_{s\to\infty} \left(1 - \frac{1}{s}\right) = 1,$$

and for $t > 1$ the same computation on $[t,s]$ gives $\int \chi_{[t,\infty)} q\, d\lambda = \lim_{s\to\infty}(1/t - 1/s) = 1/t$. Hence $\int |h|\, d\lambda \le 1$ by 3.8, and thus $h \in \mathcal{L}^1(\lambda)$ and $\chi_{(t,\infty)} h \in \mathcal{L}^1(\lambda)$; let $H = \int h\, d\lambda$. Thus the bound of an integral by the integral of the absolute value (3.23), with 3.8, gives $\left|\int \chi_{(t,\infty)} h\, d\lambda\right| \le \int \chi_{(t,\infty)} |h|\, d\lambda \le \int \chi_{[t,\infty)} q\, d\lambda = 1/t$.

Suppose $t > 1$. The function $u(x) = -(\cos x)/x$ satisfies $u'(x) = (\sin x)/x + (\cos x)/x^2 = g(x) + h(x)$ on $[1,t]$, where g and h are continuous. Thus the fundamental theorem of calculus and Lemma 1 give

$$\int \chi_{[1,t]} g\, d\lambda = \int_1^t g(x)\, dx = u(t) - u(1) - \int_1^t h(x)\, dx = \cos 1 - \frac{\cos t}{t} - \int \chi_{[1,t]} h\, d\lambda.$$

Now $\chi_{(0,t)} g + \chi_{\{t\}} g = \chi_{(0,1)} g + \chi_{[1,t]} g$ and $h = \chi_{[1,t]} h + \chi_{(t,\infty)} h$ on $\mathbf{R}$. Each function in these two equations is in $\mathcal{L}^1(\lambda)$ by the two bounds above, and $\int \chi_{\{t\}} g\, d\lambda = 0$ because $\lambda(\{t\}) = 0$. Thus

additivity of integration (3.21), applied to both equations, gives

$$F(t) = c + \int \chi_{[1,t]} g \, d\lambda = c + \cos 1 - \frac{\cos t}{t} - H + \int \chi_{(t,\infty)} h \, d\lambda,$$

where $c = \int \chi_{(0,1)} g \, d\lambda \in \mathbf{R}$. Let $L = c + \cos 1 - H$. Because $|\cos t| \le 1$ and $\left|\int \chi_{(t,\infty)} h \, d\lambda\right| \le 1/t$, the triangle inequality gives $|F(t) - L| \le 2/t$ for all $t > 1$. Hence $\lim_{t\to\infty} \int_{(0,t)} f \, d\lambda = L$ exists in $\mathbf{R}$, as desired. □

Lemma 2. Suppose $h\colon \mathbf{R} \to [0,\infty)$ is Borel measurable, $a \in \mathbf{R}$, $s \in \mathbf{R} \setminus \{0\}$, $h_a(x) = h(x-a)$ and $h^{(s)}(x) = h(sx)$. Then

$$\int h_a \, d\lambda = \int h \, d\lambda \qquad \text{and} \qquad \int h^{(s)} \, d\lambda = \frac{1}{|s|} \int h \, d\lambda.$$

Moreover, $x \mapsto u(x-a)$ and $x \mapsto u(sx)$ are Borel measurable for every Borel measurable $u\colon \mathbf{R} \to \mathbf{R}$.

Proof. The maps $x \mapsto x-a$ and $x \mapsto sx$ are continuous, and thus by the composition of measurable functions (2.44) the functions $u(x-a)$ and $u(sx)$ are Borel measurable for every Borel measurable u, and $a + A$ and $s^{-1}A$ are Borel sets for every Borel set A. If $A_1, \dots, A_m$ is a $\mathcal{B}$-partition of $\mathbf{R}$, then so is $a + A_1, \dots, a + A_m$, with $\inf_{a+A_j} h_a = \inf_{A_j} h$ and $\lambda(a + A_j) = \lambda(A_j)$ because outer measure is translation invariant (2.7); every $\mathcal{B}$-partition of $\mathbf{R}$ is such a translate, and thus the lower Lebesgue sums (3.2) of h_a and of h have the same supremum, which is the first equation. Scaling by s the open intervals of a cover of A gives a cover of sA with lengths multiplied by $|s|$, and thus $|sA| \le |s|\,|A|$ by the definition of outer measure (2.2); applying this to sA and $1/s$ gives $\lambda(sA) = |s|\,\lambda(A)$, and the partitions $s^{-1}A_1, \dots, s^{-1}A_m$ give the second equation in the same way. □

Exercise 3.2 (MIRA 3B.16). Suppose $f \in \mathcal{L}^1(\mathbf{R})$.

(a) For $t \in \mathbf{R}$, define $f_t\colon \mathbf{R} \to \mathbf{R}$ by $f_t(x) = f(x-t)$. Prove that

$$\lim_{t\to 0} \|f - f_t\|_1 = 0.$$

(b) For $t > 0$, define $f_t\colon \mathbf{R} \to \mathbf{R}$ by $f_t(x) = f(tx)$. Prove that

$$\lim_{t\to 1} \|f - f_t\|_1 = 0.$$

Solution 3.2. Here λ is Lebesgue measure on the Borel subsets of $\mathbf{R}$, as in the definition of $\mathcal{L}^1(\mathbf{R})$ (3.45); thus f is Borel measurable and $\|f\|_1 = \int |f| \, d\lambda < \infty$.

(a) Suppose $\varepsilon > 0$. Approximation by continuous functions (3.48) gives a continuous $g\colon \mathbf{R} \to \mathbf{R}$ with $\|f - g\|_1 < \varepsilon/3$ and $\{x \in \mathbf{R} : g(x) \ne 0\}$ bounded, hence contained in $[-M, M]$ for some $M > 0$. Because g is continuous, $g - f$ is Borel measurable with $\|g - f\|_1 < \varepsilon/3$, and thus $g = f + (g - f) \in \mathcal{L}^1(\mathbf{R})$ by the properties of the $\mathcal{L}^1$-norm (3.43). Let $\eta = \varepsilon/\big(3(2M+2)\big)$. Because g is continuous on the closed bounded set $[-M-1, M+1]$ and continuous functions on closed bounded sets are uniformly continuous, there exists $\delta \in (0, 1]$ such that $|g(x) - g(y)| < \eta$ for all $x, y \in [-M-1, M+1]$ with $|x - y| < \delta$. If $x, y \in \mathbf{R}$ with $|x - y| < \delta$ do not both lie in $[-M-1, M+1]$, say $|x| > M+1$, then $|y| \ge |x| - |x-y| > M$ by the triangle inequality, and thus $g(x) = g(y) = 0$. Thus $|g(x) - g(y)| < \eta$ for all $x, y \in \mathbf{R}$ with $|x - y| < \delta$.

Now suppose $t \in \mathbf{R}$ with $|t| < \delta$. Taking $y = x - t$ above gives $|g(x) - g_t(x)| < \eta$ for all $x \in \mathbf{R}$, and $g(x) = g_t(x) = 0$ when $|x| > M + 1$. Hence $|g - g_t| \le \eta\, \chi_{[-M-1,M+1]}$. Thus integration is order preserving (3.8) and the integral of a simple function (3.7) give $\|g - g_t\|_1 \le \eta\, \lambda([-M-1, M+1]) = \eta(2M+2) = \varepsilon/3$. Next, $|f_t - g_t|$, $|f_t|$ and $|g_t|$ are the translates by t of the nonnegative functions $|f - g|$, $|f|$ and $|g|$. Thus the first equation of Lemma 2 gives $\|f_t - g_t\|_1 = \|f - g\|_1 < \varepsilon/3$, $\|f_t\|_1 = \|f\|_1 < \infty$ and $\|g_t\|_1 = \|g\|_1 < \infty$; in particular, $f_t, g_t \in \mathcal{L}^1(\mathbf{R})$. Now $f - f_t = (f - g) + (g - g_t) + (g_t - f_t)$, and thus 3.43 gives

$$\|f - f_t\|_1 \le \|f - g\|_1 + \|g - g_t\|_1 + \|f_t - g_t\|_1 < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon$$

for all $t \in \mathbf{R}$ with $|t| < \delta$. Thus $\lim_{t\to 0} \|f - f_t\|_1 = 0$.

(b) Suppose $\varepsilon > 0$. By 3.48 there exists a continuous $g\colon \mathbf{R} \to \mathbf{R}$ with $\|f - g\|_1 < \varepsilon/6$ and $\{x \in \mathbf{R} : g(x) \ne 0\} \subseteq [-M, M]$ for some $M > 0$. As in (a), $g \in \mathcal{L}^1(\mathbf{R})$. Let $\eta = \varepsilon/(12M)$. Because g is continuous on the closed bounded set $[-3M, 3M]$ and continuous functions on closed bounded sets are uniformly continuous, there exists $\delta > 0$ such that $|g(x) - g(y)| < \eta$ for all $x, y \in [-3M, 3M]$ with $|x - y| < \delta$. Let $\rho = \min\{\frac{1}{2}, \delta/(2M)\}$, and suppose $t > 0$ with $|t - 1| < \rho$; then $\frac{1}{2} < t < \frac{3}{2}$. If $x \in \mathbf{R}$ and $|x| \le 2M$, then $x, tx \in [-3M, 3M]$ because $t < \frac{3}{2}$, and $|x - tx| = |x|\,|1 - t| < 2M\rho \le \delta$, and thus $|g(x) - g_t(x)| < \eta$; if $|x| > 2M$, then $|x| > M$ and $|tx| > M$, and thus $g(x) = g_t(x) = 0$. Hence $|g - g_t| \le \eta\, \chi_{[-2M,2M]}$. Thus 3.8 and 3.7 give $\|g - g_t\|_1 \le \eta\, \lambda([-2M, 2M]) = 4M\eta = \varepsilon/3$. Because $|f_t - g_t| = |f - g|^{(t)}$ and $1/t < 2$, the second equation of Lemma 2, applied to the nonnegative function $|f - g|$, gives $\|f_t - g_t\|_1 = \frac{1}{t}\|f - g\|_1 < 2 \cdot \frac{\varepsilon}{6} = \frac{\varepsilon}{3}$; applied to $|f|$ and $|g|$ it gives $\|f_t\|_1 = \|f\|_1/t < \infty$ and $\|g_t\|_1 = \|g\|_1/t < \infty$, and thus $f_t, g_t \in \mathcal{L}^1(\mathbf{R})$. Thus 3.43 gives

$$\|f - f_t\|_1 \le \|f - g\|_1 + \|g - g_t\|_1 + \|f_t - g_t\|_1 < \frac{\varepsilon}{6} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} < \varepsilon$$

for all $t > 0$ with $|t - 1| < \rho$. Hence $\lim_{t\to 1} \|f - f_t\|_1 = 0$. □

Exercise 3.3 (MIRA 4A.1). Suppose $(X, \mathcal{S}, \mu)$ is a measure space and $h\colon X \to \mathbf{R}$ is an $\mathcal{S}$-measurable function. Prove that

$$\mu(\{x \in X : |h(x)| \ge c\}) \le \frac{1}{c^p} \int |h|^p \, d\mu$$

for all positive numbers c and p.

Solution 3.3. Suppose $c > 0$ and $p > 0$, and let $A = \{x \in X : |h(x)| \ge c\}$. The function $|h|^p$ is $\mathcal{S}$-measurable by composition of measurable functions (2.44), being the composition of h with the continuous function $y \mapsto |y|^p$. Because $y \mapsto y^p$ is strictly increasing on $[0, \infty)$, we have $A = \{x \in X : |h(x)|^p \ge c^p\}$, the inverse image of the closed set $[c^p, \infty)$ under $|h|^p$. Thus $A \in \mathcal{S}$ and $c^p \chi_A \le |h|^p$ on X. Hence the integral of a simple function (3.7) and order preservation of integration (3.8) give

$$c^p \mu(A) = \int c^p \chi_A \, d\mu \le \int |h|^p \, d\mu.$$

Dividing by $c^p > 0$ gives the desired inequality. □

Exercise 3.4 (MIRA 4A.2). Suppose $(X, \mathcal{S}, \mu)$ is a measure space with $\mu(X) = 1$ and $h \in \mathcal{L}^1(\mu)$. Prove that

$$\mu\Big(\Big\{x \in X : \Big|h(x) - \int h\,d\mu\Big| \geq c\Big\}\Big) \leq \frac{1}{c^2}\Big(\int h^2\,d\mu - \Big(\int h\,d\mu\Big)^2\Big)$$

for all $c > 0$. [The result above is called Chebyshev's inequality; it plays an important role in probability theory. Pafnuty Chebyshev (1821–1894) was Markov's thesis advisor.]

Solution 3.4. Let $a = \int h\,d\mu$, which is a real number because $h \in \mathcal{L}^1(\mu)$, and suppose $c > 0$. The function h^2 is $\mathcal{S}$-measurable. If $\int h^2\,d\mu = \infty$, then the right side of the desired inequality is ∞ and the inequality holds. Thus assume $\int h^2\,d\mu < \infty$.

Let $k = h - a$, which is $\mathcal{S}$-measurable by algebraic operations with measurable functions (2.46(a)). Now $k^2 = (h^2 + a^2) - 2ah$ on X. Additivity of integration for nonnegative functions (3.16) and the integral of a simple function (3.7) give $\int (h^2 + a^2)\,d\mu = \int h^2\,d\mu + a^2\mu(X) = \int h^2\,d\mu + a^2 < \infty$, because $\mu(X) = 1$. Homogeneity of integration (3.20) gives $\int |2ah|\,d\mu = 2|a|\|h\|_1 < \infty$. Thus additivity of integration (3.21) and 3.20 give $\int k^2\,d\mu = \int h^2\,d\mu + a^2 - 2a\int h\,d\mu = \int h^2\,d\mu - a^2$. Hence Exercise 4A.1 applied to k with $p = 2$ gives

$$\begin{aligned}\mu\Big(\Big\{x \in X : \Big|h(x) - \int h\,d\mu\Big| \geq c\Big\}\Big) &= \mu(\{x \in X : |k(x)| \geq c\}) \\ &\leq \frac{1}{c^2}\int k^2\,d\mu = \frac{1}{c^2}\Big(\int h^2\,d\mu - \Big(\int h\,d\mu\Big)^2\Big). \qquad \square\end{aligned}$$

Exercise 3.5 (MIRA 4A.10). Prove or give a counterexample: If $h\colon \mathbf{R} \to [0, \infty)$ is an increasing function, then h^* is an increasing function.

Solution 3.5. The statement is true. Suppose $h\colon \mathbf{R} \to [0, \infty)$ is increasing. Then h is Borel measurable because every increasing function is Borel measurable (2.43), and $|h| = h$. Thus we take λ to be Lebesgue measure on the Borel sets, as the definition of $\mathcal{L}^1(\mathbf{R})$ (3.45) allows and as in Exercise 3B.16.

Suppose $b < b'$, let $s = b' - b > 0$, and suppose $t > 0$. Let $f = \chi_{(b'-t,b'+t)}h$ and $f' = \chi_{(b-t,b+t)}h$, nonnegative Borel measurable functions with $\int f\,d\lambda = \int_{b'-t}^{b'+t} h$ and $\int f'\,d\lambda = \int_{b-t}^{b+t} h$ by the definition of $\int_a^b$ (3.39). Let $g(x) = f(x + s)$ for $x \in \mathbf{R}$, also nonnegative and Borel measurable. By Lemma 2, the integral of a nonnegative Borel measurable function is unchanged by translation; thus $\int g\,d\lambda = \int f\,d\lambda$. If $x \in \mathbf{R}$, then $x + s \in (b' - t, b' + t)$ if and only if $x \in (b - t, b + t)$, and $h(x) \leq h(x+s)$ because h is increasing; thus $f'(x) = \chi_{(b-t,b+t)}(x)h(x) \leq \chi_{(b-t,b+t)}(x)h(x+s) = g(x)$. Hence order preservation of integration (3.8) gives

$$\frac{1}{2t}\int_{b-t}^{b+t} h = \frac{1}{2t}\int f'\,d\lambda \leq \frac{1}{2t}\int g\,d\lambda = \frac{1}{2t}\int f\,d\lambda = \frac{1}{2t}\int_{b'-t}^{b'+t} h \leq h^*(b').$$

Taking the supremum over $t > 0$ gives $h^*(b) \leq h^*(b')$ by the definition of the Hardy–Littlewood maximal function (4.6). Thus h^* is increasing. $\square$

For $f \in \mathcal{L}^1(\mathbf{R})$ and I an interval of $\mathbf{R}$ with $0 < |I| < \infty$, let f_I denote the average of f on I. In other words, $f_I = \frac{1}{|I|}\int_I f$.

Lemma 3. Suppose $u < v$ and I is an interval with $(u, v) \subseteq I \subseteq [u, v]$. Then $|I| = v - u$, and $\int_I g = \int_u^v g$ for every Lebesgue measurable $g\colon \mathbf{R} \to [0, \infty)$. If $f \in \mathcal{L}^1(\mathbf{R})$ and $b \in \mathbf{R}$, then $\int_I |f - f_I| \le 2 \int_I |f - f(b)|$.

Proof. The set I is a Borel set, being the union of (u, v) and a subset of $\{u, v\}$. Because outer measure preserves order (2.5) and the outer measure of a closed interval is its length (2.14), we have $|I| \le v - u$. Finite sets have outer measure 0 (Example 2.3), and thus subadditivity of outer measure (2.8) gives $v - u = |[u, v]| \le |I| + |\{u, v\}| = |I|$. Hence $|I| = v - u$. Now suppose $g\colon \mathbf{R} \to [0, \infty)$ is Lebesgue measurable, and let $E = I \setminus (u, v)$, a subset of $\{u, v\}$. Then $|E| = 0$ and $g \le M$ on E, where $M = \max\{g(u), g(v)\}$. Order preservation of integration (3.8) and the integral of a simple function (3.7) give $0 \le \int_E g \le M|E| = 0$. Because $\chi_I g = \chi_{(u,v)} g + \chi_E g$, additivity of integration for nonnegative functions (3.16) and the definition of $\int_a^b$ (3.39) give $\int_I g = \int_{(u,v)} g + \int_E g = \int_u^v g$.

Now suppose $f \in \mathcal{L}^1(\mathbf{R})$ and $b \in \mathbf{R}$. Because $|\chi_I f| \le |f|$, 3.8 gives $\chi_I f \in \mathcal{L}^1(\mathbf{R})$, and thus f_I is defined. By 3.7 and homogeneity of integration (3.20), $\int_I f(b) = f(b)|I|$. Thus additivity of integration (3.21) and 3.20 give $\int_I \big(f(b) - f\big) = |I|\big(f(b) - f_I\big)$. Because the absolute value of an integral is at most the integral of the absolute value (3.23), we have $|I||f(b) - f_I| \le \int_I |f - f(b)|$. By the triangle inequality, $|f - f_I| \le |f - f(b)| + |f(b) - f_I|$ on $\mathbf{R}$. Multiplying by χ_I and integrating, 3.8 and 3.16 give $\int_I |f - f_I| \le \int_I |f - f(b)| + |f(b) - f_I||I| \le 2 \int_I |f - f(b)|$, where the last step is the inequality above. □

Exercise 3.6 (MIRA 4B.1). Suppose $f \in \mathcal{L}^1(\mathbf{R})$. Prove that

$$\lim_{t \downarrow 0} \frac{1}{2t} \int_{b-t}^{b+t} |f - f_{[b-t,b+t]}| = 0$$

for almost every $b \in \mathbf{R}$.

Solution 3.6. Suppose $b \in \mathbf{R}$ and $t > 0$, and let $I = [b - t, b + t]$; then $|I| = 2t$ by Lemma 3, and

$$0 \le \frac{1}{2t} \int_{b-t}^{b+t} |f - f_{[b-t,b+t]}| \le \frac{1}{2t} \int_I |f - f_I| \le \frac{1}{t} \int_I |f - f(b)| = 2 \cdot \frac{1}{2t} \int_{b-t}^{b+t} |f - f(b)|,$$

where the second inequality and the last equation hold because endpoints do not change the integral (Lemma 3) and the third inequality is the inequality of Lemma 3. The right side tends to 0 as $t \downarrow 0$ for almost every $b \in \mathbf{R}$ by the first version of the Lebesgue Differentiation Theorem (4.10), completing the proof. □

Exercise 3.7 (MIRA 4B.2). Suppose $f \in \mathcal{L}^1(\mathbf{R})$. Prove that

$$\lim_{t \downarrow 0} \sup \left\{ \frac{1}{|I|} \int_I |f - f_I| : I \text{ is an interval of length } t \text{ containing } b \right\} = 0$$

for almost every $b \in \mathbf{R}$.

Solution 3.7. Suppose $b \in \mathbf{R}$, $t > 0$, and I is an interval of length t containing b. Because $0 < |I| < \infty$, the interval I is bounded and contains more than one point, and thus has endpoints $u < v$ in $\mathbf{R}$ with $(u, v) \subseteq I \subseteq [u, v]$; by Lemma 3, $v - u = |I| = t$. Because $u \le b \le v$, we have $u \ge b - t$ and $v \le b + t$; thus $I \subseteq [b - t, b + t]$ and $\chi_I |f - f(b)| \le \chi_{[b-t,b+t]} |f - f(b)|$ on $\mathbf{R}$.

By Lemma 3, $\int_I |f - f_I| \le 2\int_I |f - f(b)|$ and $\int_{[b-t,b+t]} |f - f(b)| = \int_{b-t}^{b+t} |f - f(b)|$. Thus order preservation of integration (3.8) gives

$$0 \le \frac{1}{|I|}\int_I |f - f_I| \le \frac{2}{t}\int_{[b-t,b+t]} |f - f(b)| = 4 \cdot \frac{1}{2t}\int_{b-t}^{b+t} |f - f(b)|.$$

The right side does not depend on I. Hence the supremum in the statement lies between 0 and the right side, which tends to 0 as $t \downarrow 0$ for almost every $b \in \mathbf{R}$ by the first version of the Lebesgue Differentiation Theorem (4.10), as desired. □

Exercise 3.8 (MIRA 4B.3). Suppose $f : \mathbf{R} \to \mathbf{R}$ is a Lebesgue measurable function such that $f^2 \in \mathcal{L}^1(\mathbf{R})$. Prove that

$$\lim_{t \downarrow 0} \frac{1}{2t}\int_{b-t}^{b+t} |f - f(b)|^2 = 0$$

for almost every $b \in \mathbf{R}$.

Solution 3.8. For $n \in \mathbf{Z}^+$, let $f_n = \chi_{(-n,n)} f$, which is Lebesgue measurable. Because $0 \le (|f|-1)^2$, we have $|f| \le \frac{1}{2}(1 + f^2)$ on $\mathbf{R}$, and thus $|f_n| \le \frac{1}{2}\chi_{(-n,n)} + \frac{1}{2}f^2$. Order preservation of integration (3.8) and additivity of integration for nonnegative functions (3.16) give $\int |f_n|\, d\lambda \le \int \frac{1}{2}\chi_{(-n,n)}\, d\lambda + \int \frac{1}{2}f^2\, d\lambda$. The first integral on the right is $\frac{1}{2}|(-n,n)| = n$ by the integral of a simple function (3.7), and the second is $\frac{1}{2}\int f^2\, d\lambda < \infty$ by homogeneity of integration (3.20). Thus $f_n \in \mathcal{L}^1(\mathbf{R})$. The result that an $\mathcal{L}^1(\mathbf{R})$ function equals its local average almost everywhere (4.21), applied to f^2 and to each f_n, gives sets $A_0, A_1, A_2, \ldots$ with $|A_n| = 0$ such that $f(b)^2 = \lim_{t\downarrow 0} \frac{1}{2t}\int_{b-t}^{b+t} f^2$ for every $b \in \mathbf{R} \setminus A_0$ and $f_n(b) = \lim_{t\downarrow 0} \frac{1}{2t}\int_{b-t}^{b+t} f_n$ for every $b \in \mathbf{R} \setminus A_n$. Let $A = \bigcup_{n=0}^{\infty} A_n$; then $|A| = 0$ by countable subadditivity of outer measure (2.8).

Suppose $b \in \mathbf{R} \setminus A$. Let $n \in \mathbf{Z}^+$ be such that $n > |b|$; then $b \in (-n,n)$ and $f_n(b) = f(b)$. Suppose $0 < t < n - |b|$, and let $J = (b - t, b + t)$. Then $J \subseteq (-n,n)$, which implies that $\chi_J f_n = \chi_J f$, and $|J| = 2t$. On $\mathbf{R}$ we have $|f - f(b)|^2 = f^2 - 2f(b)f + f(b)^2$, and thus

$$\chi_J |f - f(b)|^2 = \chi_J f^2 - 2f(b)\,\chi_J f_n + f(b)^2 \chi_J.$$

The three functions on the right are Lebesgue measurable and in $\mathcal{L}^1(\mathbf{R})$: $|\chi_J f^2| \le f^2$ and $|\chi_J f_n| \le |f_n|$ on $\mathbf{R}$, which gives finite integrals by 3.8, and $\int f(b)^2 \chi_J\, d\lambda = 2tf(b)^2$ by 3.7. Hence additivity of integration (3.21) and 3.20 give

$$\int \chi_J |f - f(b)|^2\, d\lambda = \int \chi_J f^2\, d\lambda - 2f(b)\int \chi_J f_n\, d\lambda + 2tf(b)^2.$$

Dividing by $2t$ and using the definition of $\int_a^b$ (3.39), we have

$$\frac{1}{2t}\int_{b-t}^{b+t} |f - f(b)|^2 = \frac{1}{2t}\int_{b-t}^{b+t} f^2 - 2f(b) \cdot \frac{1}{2t}\int_{b-t}^{b+t} f_n + f(b)^2$$

for every $t \in (0, n - |b|)$. As $t \downarrow 0$, the first term on the right tends to $f(b)^2$ because $b \notin A_0$, and the second tends to $-2f(b)f_n(b) = -2f(b)^2$ because $b \notin A_n$. Thus the right side tends to $f(b)^2 - 2f(b)^2 + f(b)^2 = 0$. Because $|A| = 0$, this proves the desired limit for almost every $b \in \mathbf{R}$. □

Exercise 3.9 (MIRA 4B.4). Prove that the Lebesgue Differentiation Theorem (4.19) still holds if the hypothesis that $\int_{-\infty}^{\infty}|f|<\infty$ is weakened to the requirement that $\int_{-\infty}^{x}|f|<\infty$ for all $x\in\mathbf{R}$.

Solution 3.9. Suppose $f\colon\mathbf{R}\to\mathbf{R}$ is Lebesgue measurable and $\int_{-\infty}^{x}|f|<\infty$ for every $x\in\mathbf{R}$. For $x\in\mathbf{R}$, the function $\chi_{(-\infty,x)}f$ is Lebesgue measurable and $\int|\chi_{(-\infty,x)}f|\,d\lambda=\int_{-\infty}^{x}|f|<\infty$ by the definition of $\int_a^b$ (3.39); thus $\chi_{(-\infty,x)}f\in\mathcal{L}^1(\mathbf{R})$, and $g(x)=\int_{-\infty}^{x}f$ defines a function $g\colon\mathbf{R}\to\mathbf{R}$. For $n\in\mathbf{Z}^+$, let $f_n=\chi_{(-\infty,n)}f$, which is in $\mathcal{L}^1(\mathbf{R})$ by the case $x=n$, and define $g_n\colon\mathbf{R}\to\mathbf{R}$ by $g_n(x)=\int_{-\infty}^{x}f_n$. If $x\le n$, then $\chi_{(-\infty,x)}f_n=\chi_{(-\infty,x)}f$, and thus $g_n(x)=g(x)$. By the second version of the Lebesgue Differentiation Theorem (4.19), applied to f_n, there is a set A_n with $|A_n|=0$ such that $g_n'(b)=f_n(b)$ for every $b\in\mathbf{R}\setminus A_n$. Countable subadditivity of outer measure (2.8) gives $|A|=0$ for $A=\bigcup_{n=1}^{\infty}A_n$.

Now suppose $b\in\mathbf{R}\setminus A$. Let $n\in\mathbf{Z}^+$ be such that $n>b$. Then $g=g_n$ on the open interval $(-\infty,n)$, which contains b, and thus g and g_n have the same difference quotients $\big(g(b+t)-g(b)\big)/t$ for all t with $0<|t|<n-b$; hence by the definition of the derivative (4.16), g is differentiable at b if and only if g_n is, with the same derivative. Because $b\notin A_n$, g_n is differentiable at b with $g_n'(b)=f_n(b)$, and $f_n(b)=f(b)$ because $b<n$. Hence g is differentiable at b with $g'(b)=f(b)$ for every $b\in\mathbf{R}\setminus A$, and thus for almost every $b\in\mathbf{R}$, as desired. □

Exercise 3.10 (MIRA 4B.5). Suppose $f\colon\mathbf{R}\to\mathbf{R}$ is a Lebesgue measurable function. Prove that

$$|f(b)|\le f^*(b)$$

for almost every $b\in\mathbf{R}$.

Solution 3.10. For $n\in\mathbf{Z}^+$, let $E_n=\{x\in(-n,n):|f(x)|\le n\}$, a Lebesgue measurable set, and let $f_n=\chi_{E_n}f$, which is Lebesgue measurable, with $|f_n|\le n\chi_{(-n,n)}$ on $\mathbf{R}$. Thus order preservation of integration (3.8) and the integral of a simple function (3.7) give $\int|f_n|\,d\lambda\le n|(-n,n)|=2n^2<\infty$; hence $f_n\in\mathcal{L}^1(\mathbf{R})$. Because an $\mathcal{L}^1(\mathbf{R})$ function equals its local average almost everywhere (4.21), there is a set A_n with $|A_n|=0$ such that $f_n(b)=\lim_{t\downarrow0}\frac{1}{2t}\int_{b-t}^{b+t}f_n$ for every $b\in\mathbf{R}\setminus A_n$. Let $A=A_1\cup A_2\cup\cdots$. Then $|A|\le\sum_{n=1}^{\infty}|A_n|=0$ by countable subadditivity of outer measure (2.8).

Suppose $b\in\mathbf{R}\setminus A$. There exists $n\in\mathbf{Z}^+$ such that $n>|b|$ and $n>|f(b)|$. Then $b\in E_n$, and thus $f_n(b)=f(b)$. Suppose $t>0$. Because $f_n\in\mathcal{L}^1(\mathbf{R})$, the absolute value of an integral is at most the integral of the absolute value (3.23), and $|f_n|\le|f|$ on $\mathbf{R}$. Thus 3.8 and the definition of f^* (4.6) give

$$\Big|\frac{1}{2t}\int_{b-t}^{b+t}f_n\Big|\le\frac{1}{2t}\int_{b-t}^{b+t}|f_n|\le\frac{1}{2t}\int_{b-t}^{b+t}|f|\le f^*(b).$$

Because $b\notin A_n$, letting $t\downarrow0$ gives $|f(b)|=|f_n(b)|\le f^*(b)$. This holds for every $b\in\mathbf{R}\setminus A$, and thus for almost every $b\in\mathbf{R}$. □